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Multi-hazard risk mapping using geospatial techniques in coastal Bangladesh: implications for resilience

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ABSTRACT

Coastal areas play a vital role in supporting livelihoods, resources, and economic activities, yet in Bangladesh they are highly vulnerable to climate-induced hazards such as cyclones, floods, salinity intrusion, and erosion. These hazards are expected to intensify under changing climatic conditions, increasing risks to coastal communities. This study focuses on Kalapara Upazila in the southern coastal region of Bangladesh and develops an integrated multi-hazard risk map using geospatial techniques. A total of 34 spatial criteria were analyzed under three key components: hazard, vulnerability, and mitigation capacity. Each criterion was transformed into spatial layers and weighted using the Analytical Hierarchy Process (AHP). The resulting component indices were integrated through a weighted overlay to generate the final risk map. The findings reveal that 16.08% of the area falls within very high-risk zones and 34.03% within high-risk zones, mainly concentrated in the southern and southwestern parts, including Lalua, Khaprabhanga, Baliatali, Latachapli, and Mithaganj. In contrast, low and very low-risk zones cover 14.25% and 7.76% of the area, primarily in northern and central regions. The results provide valuable insights for policymakers to support targeted disaster risk reduction and sustainable coastal management.

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KEY WORDS
Multi hazard; risk assessment; GIS; remote sensing; analytical hierarchy process

1. Introduction

Coastal zones are among the world's most valuable yet hazard-prone environments, supporting substantial ecological, social, and economic activities (Ahmed et al. 2021). Owing to their proximity to the sea, high population densities, and concentration of economic assets, coastal regions are disproportionately exposed to natural hazards such as tropical cyclones, storm surges, coastal flooding, sea-level rise, and erosion (Sahoo and Bhaskaran 2018; Rangel-Buitrago et al. 2018). Approximately 70% of the global population resides in coastal areas, making these regions particularly vulnerable to disaster-related losses (Willer et al. 2025). In recent decades, the frequency and impacts of natural hazards have increased considerably, resulting in significant human and economic losses worldwide (Lincke et al. 2022; EM-DAT 2025). Moreover, climate change and rising sea levels are expected to further intensify risks in low-lying coastal regions throughout the twenty-first century (IPCC 2014; Palmer et al. 2020; Li et al. 2025). Consequently, coastal communities face growing threats to their lives, livelihoods, infrastructure, and ecosystems, highlighting the need for effective risk assessment and climate-resilient coastal management.

Bangladesh is among the world's most disaster-prone countries, with high vulnerability to natural hazards and climate change due to poverty, high population density, and limited adaptive capacity (Zaman et al. 2022; Das et al. 2024). Its coastal region comprises 19 districts, covering approximately 47,200 km^2 (32% of the country's land area) and supporting around 43.8 million people (Bangladesh Bureau of Statistics 2022). The region is frequently affected by multiple interacting hazards, including tropical cyclones, storm surges, salinity intrusion, and coastal erosion (Islam et al. 2021). Tropical cyclones are particularly destructive because Bangladesh's funnel-shaped coastline and low-lying deltaic topography amplify storm surges and coastal

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flooding (Akter and Sultana 2022; Arif et al. 2023; Rahman et al. 2024). Meanwhile, salinity intrusion, driven by sea-level rise, climate change, and excessive groundwater abstraction, threatens agriculture, freshwater resources, and livelihoods (Hasan et al. 2021; Ashrafuzzaman et al. 2022). Coastal erosion further damages settlements, infrastructure, and productive land, particularly along the southeastern coastline (Thomas et al. 2018; Abedin et al. 2019). Collectively, these interacting hazards highlight the need for integrated multi-hazard risk assessment to support disaster risk reduction and climate-resilient coastal planning.

When multiple hazards occur simultaneously, their combined effects can devastate communities and force displacement. Environmental refugees have increased over the last decade due to intensified extreme events (Islam et al. 2021). Although disasters cannot be entirely prevented, their impacts can be minimized through sound planning and risk reduction strategies. Understanding the spatial distribution of hazards, their contributing factors, and overall risk patterns is essential, often through comprehensive risk assessments (Hoque et al. 2018). These assessments involve analyzing hazard, vulnerability, and exposure data, guiding early warning systems, resource allocation, and investment decisions (Perwaiz et al. 2020). Hazards threaten life and assets, vulnerability reflects sensitivity to damage, and exposure indicates the presence of people, infrastructure, and other assets in hazard-prone areas (UNDRR 2017; Huq et al. 2020). Together, these components determine disaster risk and provide the basis for identifying priority areas for mitigation, enhancing disaster preparedness, and supporting resilience-oriented planning.

Several geospatial decision-support platforms, including RISK-GIS (Owolabi and Sajjad 2023), HAZUS (Schneider and Schauer 2006), CAPRA (Cardona et al. 2010), and RiskScape (Schmidt et al. 2011), have been developed to support disaster risk assessment. Among these approaches, remote sensing and GIS enable the acquisition, integration, and analysis of multi-source spatial data for hazard mapping and risk assessment. The Analytic Hierarchy Process (AHP), a widely used multi-criteria decision-making (MCDM) technique, facilitates the integration of hazard, vulnerability, and mitigation indicators by assigning relative weights based on expert knowledge (Hoque et al. 2021; Sarker et al. 2024). Together, GIS, remote sensing, and AHP provide a robust framework for developing spatially explicit risk models and supporting evidence-based disaster risk reduction. More recently, geospatial technologies have been increasingly applied to resilience assessment by identifying vulnerable communities, evaluating adaptive capacity, and informing climate adaptation and resilience-oriented planning. Recent studies have shown that integrating geospatial technologies with resilience assessment enhances sustainable coastal management and evidence-based disaster risk reduction (Ahmed et al. 2025; Islam 2025; Faisal et al. 2026).

Previous studies in Bangladesh have extensively applied GIS, remote sensing, and spatial modelling to assess a wide range of coastal hazards, including cyclones, storm surges, floods, salinity intrusion, coastal erosion, and coastal vulnerability (Roy and Blaschke 2015; Sarwar and Woodroffe 2013; Mohit 2018; Bera et al. 2021; Hoque et al. 2019a, 2019b). These studies have significantly advanced the understanding of coastal hazard distribution, vulnerability patterns, and disaster risk. More recent research has further emphasized resilience-oriented planning by integrating adaptive capacity, disaster preparedness, and geospatial technologies (Ahmed et al. 2025; Islam 2025). Collectively, these studies demonstrate the value of geospatial approaches as effective decision-support tools for coastal hazard assessment and sustainable disaster risk management. However, most of these studies primarily assessed hazards individually or emphasized specific aspects of disaster risk, providing limited consideration of the interactions among multiple hazards and their combined influence on overall coastal risk. Although resilience has received increasing attention in recent years, comparatively few studies have integrated hazard, vulnerability, mitigation capacity, and model validation within a single geospatial assessment framework.

Despite these advances, several important research gaps remain. Most existing studies have focused on individual hazards, specific components of risk, or broad regional-scale assessments, limiting their ability to capture the combined effects of multiple interacting coastal hazards. Furthermore, relatively few studies have integrated hazard, vulnerability, and mitigation capacity within a comprehensive GIS-based multi-hazard assessment framework at the Upazila scale, where spatially explicit information is critical for effective local disaster risk reduction and planning. In addition, the validation of composite multi-hazard risk maps has received limited attention, reducing confidence in their practical applicability. Consequently, decision-makers often lack locally validated and spatially integrated information that simultaneously considers multiple hazards together with vulnerability and mitigation capacity. Addressing these methodological limitations is essential for

improving the reliability and practical application of multi-hazard risk assessments in coastal planning and disaster risk reduction.

To address these methodological gaps, this study develops and validates an integrated GIS-, remote sensing-, and Analytic Hierarchy Process (AHP)-based multi-hazard risk assessment framework for Kalapara Upazila, one of the most hazard-prone coastal regions of southern Bangladesh. By integrating hazard, vulnerability, and mitigation capacity indicators derived from multi-source geospatial datasets, the proposed framework generates a spatially explicit multi-hazard risk map to identify priority risk zones and support evidence-based disaster risk reduction, climate change adaptation, resilience building, and sustainable coastal management. Specifically, this study aims to: (1) integrate hazard, vulnerability, and mitigation capacity indicators within a GIS-based multi-hazard assessment framework; (2) evaluate the spatial distribution of multi-hazard risk across Kalapara Upazila; and (3) validate the resulting risk map to assess its reliability and applicability for disaster risk reduction and resilience-oriented planning.

2. Materials and methods

This study employed a spatial AHP-based MCDM approach integrated with GIS and remote sensing to assess multi-hazard risk in coastal Bangladesh. AHP provides a systematic framework for weighting multiple criteria through pairwise comparisons (Dewan 2013). Figure 1 presents the overall methodological workflow, including data preparation, criteria standardization, AHP weighting, risk mapping, and model validation. Multi-hazard risk was calculated using the following equation (Hoque et al. 2019a, 2019b): $\text{Risk} = \text{Hazard} \times \text{Vulnerability} \times \text{Exposure/Mitigation Capacity}$.

2.1. Study area

This study focused on Kalapara Upazila, a coastal Upazila located along the Ganges deltaic coast of southern Bangladesh. Kalapara was selected because it represents one of the country's most hazard-prone coastal regions, where tropical cyclones, storm surges, floods, salinity intrusion, and coastal erosion frequently occur within a low-lying deltaic environment. Compared with many other coastal regions, Kalapara experiences the combined effects of multiple hazards while supporting climate-sensitive livelihoods, making it an appropriate case study for integrated multi-hazard risk assessment. Furthermore, the availability of reliable geospatial, environmental, and socio-economic datasets facilitated a comprehensive spatial analysis. Although other coastal Upazilas such as Koyra, Shyamnagar, and Dacope are also highly vulnerable to coastal hazards, Kalapara was selected because it represents a typical low-lying deltaic environment where cyclones, floods, salinity intrusion, and coastal erosion occur simultaneously. In addition, the availability of consistent environmental, socio-economic, and geospatial datasets made Kalapara a suitable case study for integrated multi-hazard risk assessment.

Kalapara Upazila comprises two municipalities (Kalapara and Kuakata), 12 union parishads, 58 mauzas, and 239 villages (Ahmed et al. 2021). It is located in the southern part of Bangladesh within Patuakhali District of the Barisal Division, between 21°48′ and 22°05′ N latitudes and 90°05′ and 90°20′ E longitudes (Figure 2). The Upazila is bounded by Amtali to the north and west, Galachipa and the Rabnabad Channel to the east, and the Bay of Bengal to the south. Its low-lying and flat terrain, ranging from 0.5 to 3 m above mean sea level, increases its susceptibility to coastal hazards (Banglapedia 2023).

The region is characterized by estuaries, tidal flats, mudflats, sandy beaches, and an extensive river network formed by the Andharmanik, Nilganj, and Dhankhali rivers. However, siltation and human interventions have caused approximately 25 km of the 40-km Andharmanik River to dry up (Ali et al. 2020). The dominant non-calcareous grey floodplain and coastal saline soils, together with increasing dry-season salinity, adversely affect agricultural productivity (Khanam et al. 2020). Climate change has intensified rainfall variability and temperature extremes, while coastal erosion, sea-level rise, storm surges, and saline water intrusion continue to threaten agriculture, freshwater resources, infrastructure, and local livelihoods (Amir and Ahmed 2013; Ahsan et al. 2016; Rahman et al. 2018).

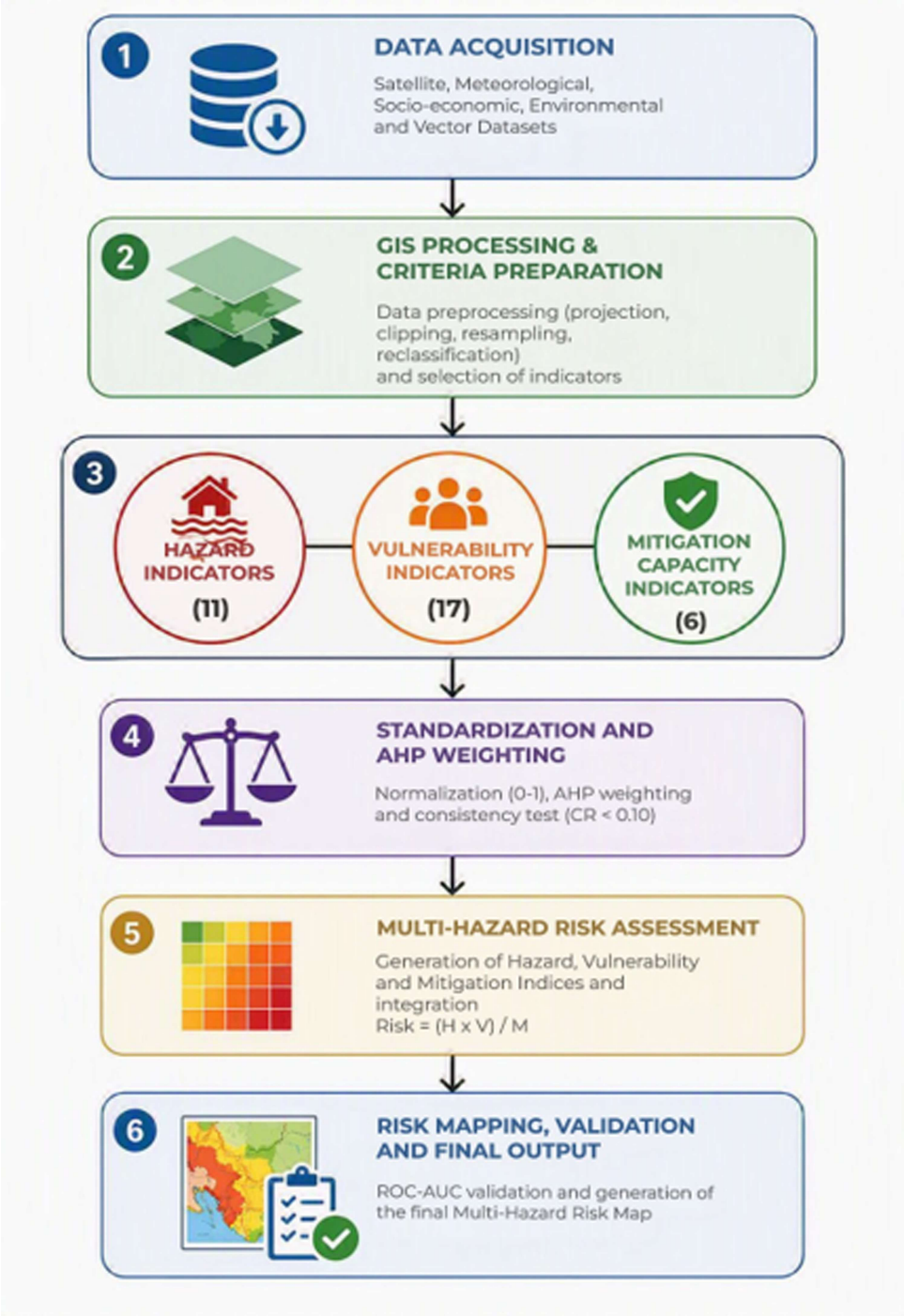


Figure 1. Methodological framework of the study showing data preparation, AHP-based weighting, risk assessment, model validation, and multi-hazard risk mapping.

2.2. Data set and sources

This study utilized approximately 34 environmental, climatic, topographic, and socio-economic parameters (Table 1), combination of climatic, environmental, social, and demographic variables. Climate-related data, including temperature, rainfall, wind speed, and evaporation, were obtained from reputable global sources such as CHIRPS, MODIS, NOAA, and the Bangladesh Meteorological Department (BMD). Localized climate and environmental indicators were sourced from national institutions such as the Bangladesh Meteorological Department (BMD). Topographic variables, including slope, elevation, aspect,

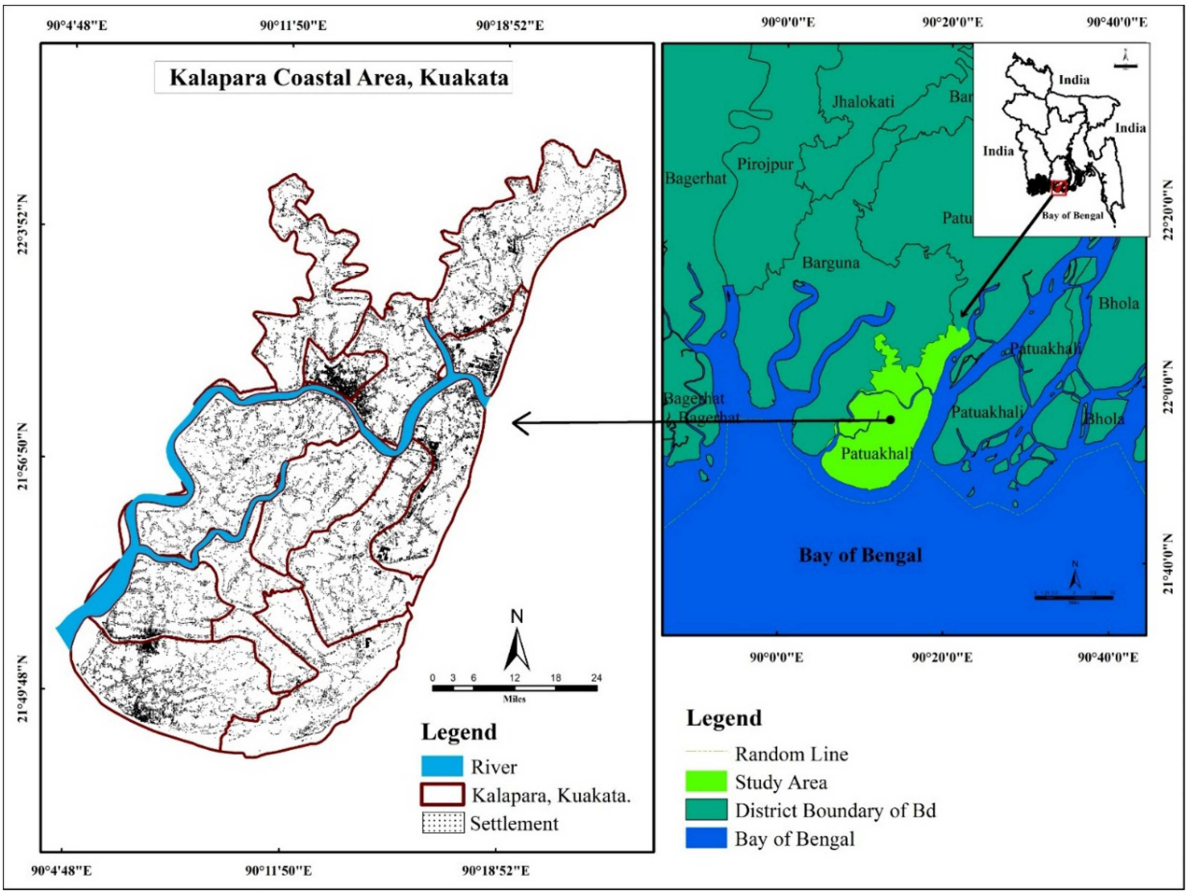


Figure 2. Geographic location of Kalapara Upazila within the Ganges deltaic coast of Bangladesh.

coastal distance, and tidal surge data, were obtained from USGS SRTM and NOAA Storm Surge. The SRTM 30 m DEM was selected because of its free availability, consistent coverage, and proven applicability in coastal hazard studies. Its accuracy was considered sufficient for deriving elevation, slope, and aspect in the predominantly rural study area. Although SRTM is technically a Digital Surface Model (DSM) and may introduce minor elevation uncertainties due to vegetation and surface objects, these effects are expected to be minimal in the present study because the study area is predominantly rural, low-lying, and contains limited dense urban development. Before terrain analysis, all raster datasets were projected, resampled to 30 m resolution, and clipped to the study area, ensuring spatial consistency for deriving elevation, slope, and aspect. Socio-economic parameters, including population density, land use, and infrastructure distribution, were drawn from datasets provided by the Bangladesh Bureau of Statistics (BBS), the Ministry of Disaster Management, and NASA Nightlight. Remote sensing and geospatial datasets, including land cover and vegetation indices, were acquired from MODIS and Landsat imagery and processed using Google Earth Engine. For dynamic online datasets (e.g. CHIRPS, MODIS, NOAA, OpenStreetMap, and Copernicus), the most recent versions available during the study period were used. Data were accessed between March and May 2024.

2.3. Risk evaluation criteria, alternatives and mapping

Drawing from the literature review, several relevant factors were selected as assessment criteria based on an extensive review of the literature, giving priority to their significance in the context of multi-hazard risk assessment. Next, we mapped all selected criteria and applied GIS and remote sensing (RS) techniques to convert them into thematic raster layers. We used a 30×30 m spatial resolution for all raster layers in the Universal Transverse Mercator (UTM) Zone 45N projection based on the World Geodetic System (WGS)

Table 1. Geospatial datasets, spatial resolutions, data sources, official web portals, and acquisition years used in this study.

Dataset type	Name	Cell size	Category	Source	Year
Static variables	Coastal elevation, slope and aspect	30 m	Topography	USGS SRTM (https://earthexplorer.usgs.gov/)	2022
	Distance to stream	30 m	Topography	Derived from SRTM using ArcGIS	2022
	Distance to coast, distance to main river	30 m	Topography	OpenStreetMap (https://www.openstreetmap.org/)	2022
	HAND (height above nearest drainage)	30 m	Topography	JRC Global HAND (https://jeodpp.jrc.ec.europa.eu/ftp/jrc-opendata/HAND/)	2021
	Cyclone track	30 m	Disaster	IBTrACS (https://www.ncei.noaa.gov/products/international-best-track-archive)	1970–2022
	Road	30 m	Infrastructure	OpenStreetMap (https://www.openstreetmap.org/)	2022
	Health facilities	30 m	Infrastructure	Ministry of Health and Family Welfare (https://www.mohfw.gov.bd/)	2021
	Shelter	30 m	Infrastructure	Ministry of Disaster Management and Relief (https://modmr.gov.bd/)	2021
	Distance to settlement	30 m	Socioeconomic	Derived in ArcGIS	2022
	Soil pH	30 m	Soil	Bangladesh Agricultural Research Council (https://barc.gov.bd/)	2021
	Clay content	30 m	Soil	Harmonized World Soil Database (https://www.fao.org/soils-portal/data-hub/soil-maps-and-databases/harmonized-world-soil-database-v12/en/)	2022
Dynamic variables	Stream density	30 m	Topography	USGS SRTM (https://earthexplorer.usgs.gov/)	2022
	LST, evaporation	30 m	Climate	MODIS (https://lpdaac.usgs.gov/)	2022
	SPI, rainfall	30 m	Climate	CHIRPS (https://www.chc.ucsb.edu/data/chirps)	2021–2022
	Tide range	30 m	Oceanographic	Bangladesh Meteorological Department (https://bmd.gov.bd/)	2022
	Storm surge	30 m	Oceanographic	NOAA (https://www.nhc.noaa.gov/)	2021
	Settlement density	30 m	Socioeconomic	NASA Black Marble Nighttime Lights (https://blackmarble.gsfc.nasa.gov/)	2022
	Population data	30 m	Socioeconomic	Bangladesh Bureau of Statistics (https://www.bbs.gov.bd/)	2022
	NDVI	30 m	Vegetation	Sentinel-2 (https://dataspace.copernicus.eu/)	2024
	Coastal vegetation	30 m	Vegetation	Landsat 8 (https://earthexplorer.usgs.gov/)	2022
	LULC	30 m	Land Use	Copernicus Global Land Cover (https://land.copernicus.eu/)	2024
	NDWI	30 m	Water Resources	Sentinel-2 (https://dataspace.copernicus.eu/)	2024
	Water pH, salinity level	30 m	Water, Soil	Soil Resource Development Institute (SRDI) (https://srdi.portal.gov.bd/)	2022

1984 datum. Regarding the classification scheme, threshold values for the majority of the criteria were adopted from published literature relevant to the study area and similar environments. For parameters without established literature-based thresholds, class intervals were generated using the Natural Breaks (Jenks) classification method in ArcGIS 10.8. The rationale and significance of the selected criteria are described below. To clearly illustrate how these selected criteria contribute to the overall risk formulation, Table 2 presents a conceptual framework linking individual parameters to the hazard, vulnerability, and mitigation components. The selected criteria were included based on their distinct conceptual roles rather than statistical independence. Although some variables (e.g. elevation, slope, HAND, and stream density) are spatially associated, each represents a different physical mechanism influencing hazard occurrence. Therefore, they were treated as conceptually independent within the AHP framework based on expert judgment and published literature.

The significance and details of the selected criteria are outlined below:

2.3.1. Criteria for hazard mapping

We have selected four major natural hazards in our study area. Flood, Cyclone, Salinity Intrusion and Erosion were considered. Most of the parameters influencing these hazards were also taken into account (Figure 3a–p) and (Figure 4a–f).

2.3.1.1. Flood hazard criteria. In this study,16 geospatial and environmental criteria were selected to assess flood hazard. These criteria were categorized under five categories ranging from Class 1 (Very Low) to Class 5 (Very High), based on thresholds derived from literature and hydrological reasoning (Table 2). Below is a description of each criterion, focusing on the rationale behind very high and very low hazard classifications.

Table 2. Conceptual framework linking selected parameters to the mathematical formulation of overall multi-hazard risk.

Risk component	Sub-component/ hazard type	Selected parameters (criteria)	Mathematical contribution to overall risk
Hazard (H)	Flood	Distance to Coast, Clay Content, Coastal DEM, Coastal Slope, Stream Density, Distance to Stream, Evaporation, Coastal Flood, LST, HAND, River Proximity, NDWI, SPI, Rainfall, Aspect.	Integrated via AHP weights to generate Hazard Index (H)
	Cyclone	Distance to Coast, Coastal DEM, Coastal Slope, Coastal Aspect, Wind Speed, Stream Density, Distance to Stream, Evaporation, LST, River Proximity, Cyclone Track, SPI, Rainfall.	
	Salinity intrusion	Coastal Aspect, Coastal Slope, Stream Density, LST, Cyclone Track, Distance to Coast, Distance to Stream, Evaporation, NDWI, Water pH, Soil pH, Coastal DEM, Main River Proximity, Tide Range, Storm Surge, Salinity Level, Rainfall.	
	Erosion	Stream Density, LST, Coastal Aspect, Coastal Slope, Cyclone Track, Distance to Coast, Distance to Stream, Evaporation, NDWI, Water pH, Soil pH, Coastal DEM, Tide Range, Main River Proximity, Storm Surge, Salinity Level, Rainfall, Clay Content, Sand Dragging.	
Vulnerability (V)	Social vulnerability	Population Density, Literacy Rate, Population Hotspot, Female Population, Dependent Population, Dependent Agriculture.	Integrated via AHP weights to generate Vulnerability Index (V)
	Physical vulnerability	Settlement Density, NDVI, Distance to Settlement, Coastal Vegetation, LULC.	
Mitigation capacity (M)	Mitigation capacity	Cyclone Centre Proximity, Health Centre Proximity, Road Distance, Distance to Coastal Forest.	Integrated via AHP weights to generate Mitigation Index (M)
Overall multi-hazard risk (R)	Final risk formulation	Hazard (H), Vulnerability (V), and Mitigation Capacity (M)	Risk = (Hazard × Vulnerability)/Mitigation Capacity

The Distance to Coast represents the horizontal distance from a specific location to the nearest coastal boundary, significantly influencing flood susceptibility in deltaic and low-lying coastal areas (Table 3). Areas less than 7 km from the coast are highly vulnerable to storm surges, tidal flooding, and cyclone induced inundation, especially in the Bay of Bengal region (Dasgupta et al. 2014).

In contrast, locations more than 28 km inland generally remain less affected by such hazards. Clay content, representing soil particles smaller than 0.002 mm, plays a key role in flood dynamics. High clay content (29%–39%) reduces permeability, limits infiltration, and increases surface runoff, thereby elevating flood risk, whereas low clay content (0%–5%) enhances infiltration and reduces water accumulation (Table 3) (Dewan 2013; Hoque et al. 2019a). Elevation derived from the Coastal Digital Elevation Model (DEM) is also critical; low-lying areas below 2 m are highly susceptible to coastal and pluvial flooding, while areas above 11 m benefit from improved drainage and lower risk (Dewan 2013; Hoque et al. 2019a).

Slope gradient further influences flood behaviour, as gentle slopes (<0.93°) promote water accumulation and prolonged inundation, whereas steeper slopes (>5.34°) facilitate faster runoff and reduced flood retention. Stream density, indicating total stream length per unit area, reflects drainage efficiency; high density (>1.5 km/km^2) suggests increased flood susceptibility due to concentrated flow paths, while low density (<0.35 km/km^2) is associated with better drainage and lower risk (Table 3) (Hoque et al. 2019a).

Proximity to streams and rivers increases flood vulnerability. Areas within 250 m of streams or 697.46 m of rivers are considered very high risk due to overflow potential, while those farther away face lower exposure. Evaporation also affects flood persistence—low rates (<0.75 mm/day) prolong flooding, whereas high rates (>4.95 mm/day) promote rapid drying (Table 3). Coastal flood extent serves as a historical indicator, with areas exceeding 22% past flood coverage being highly vulnerable. Land Surface Temperature (LST) is inversely related to moisture; low LST (<24 °C) indicates waterlogged conditions, while high LST (>33°C) reflects drier areas (Hoque et al. 2019a).

Height Above Nearest Drainage (HAND) improves flood assessment by representing elevation relative to drainage; areas with HAND <2.2 m are highly flood-prone, while those >14.51 m are relatively safe. The Normalized Difference Water Index (NDWI) identifies surface water, where values >0.20 indicate saturated zones and high risk, and values <−0.10 indicate dry conditions (Table 3). The Standardized

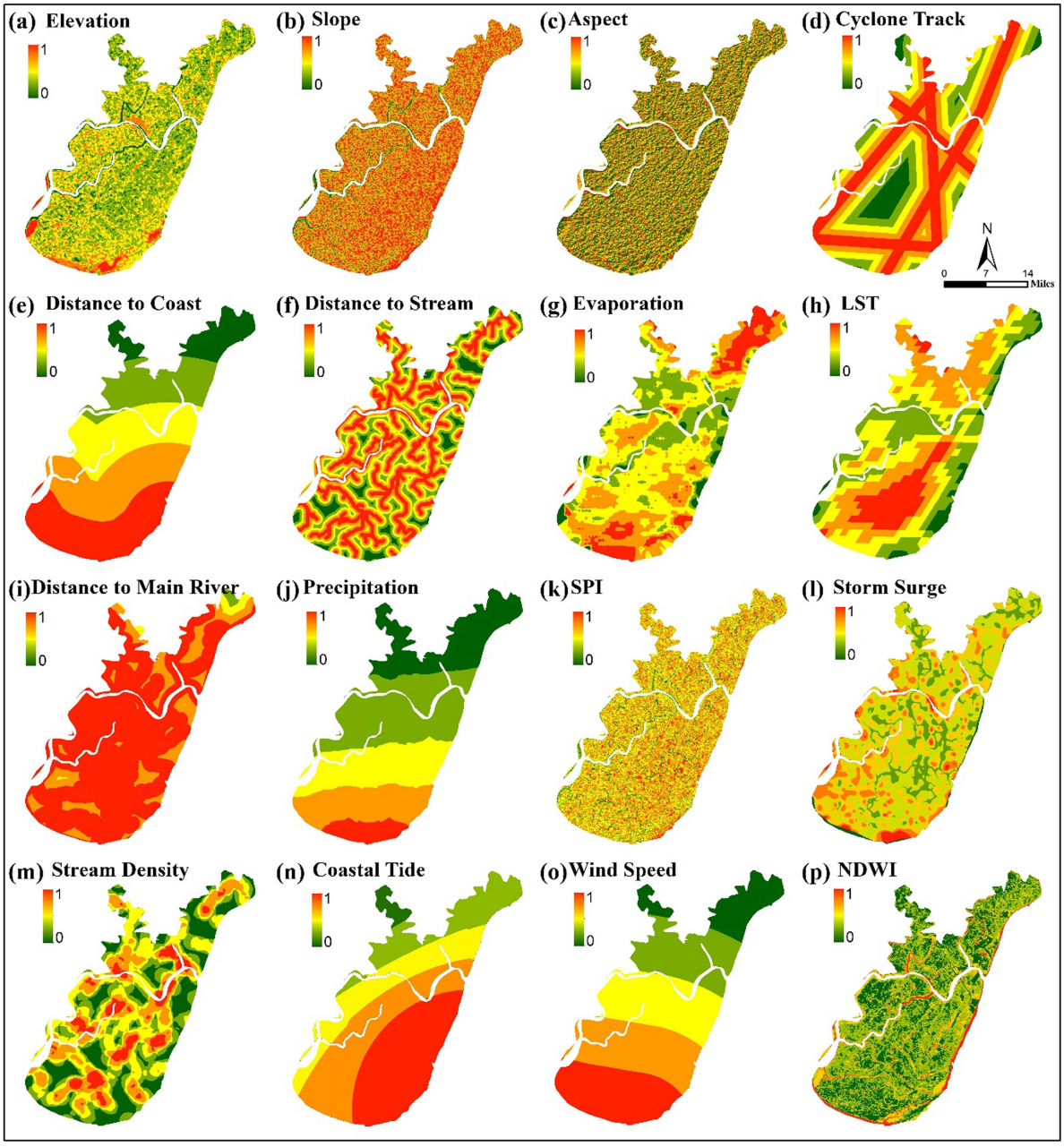


Figure 3. Hazard-related criteria used for multi-hazard assessment: (a) elevation, (b) slope, (c) aspect, (d) cyclone track, (e) distance to coast, (f) distance to stream, (g) evaporation, (h) land surface temperature (LST), (i) distance to main river, (j) precipitation, (k) SPI, (l) storm surge, (m) stream density, (n) coastal tide range, (o) wind speed, and (p) NDWI.

Precipitation Index (SPI) captures precipitation anomalies, where positive SPI values indicate wetter-than-normal conditions that increase soil moisture and flood potential; therefore, SPI has also been widely used for flood risk assessment and monitoring (Olanrewaju and Reddy 2022). Although indicators like rainfall intensity or antecedent precipitation index are common, SPI was preferred for its standardized, statistically robust measurement of precipitation anomalies. It better captures cumulative rainfall deviations over time, making it highly suitable for regional flood assessment (Olanrewaju and Reddy 2022). Annual rainfall further influences flood risk, with higher totals (>2515.17 mm) increasing susceptibility. Aspect also plays a role, as north-facing slopes tend to retain moisture, while south-east facing slopes dry faster, reducing flood risk.

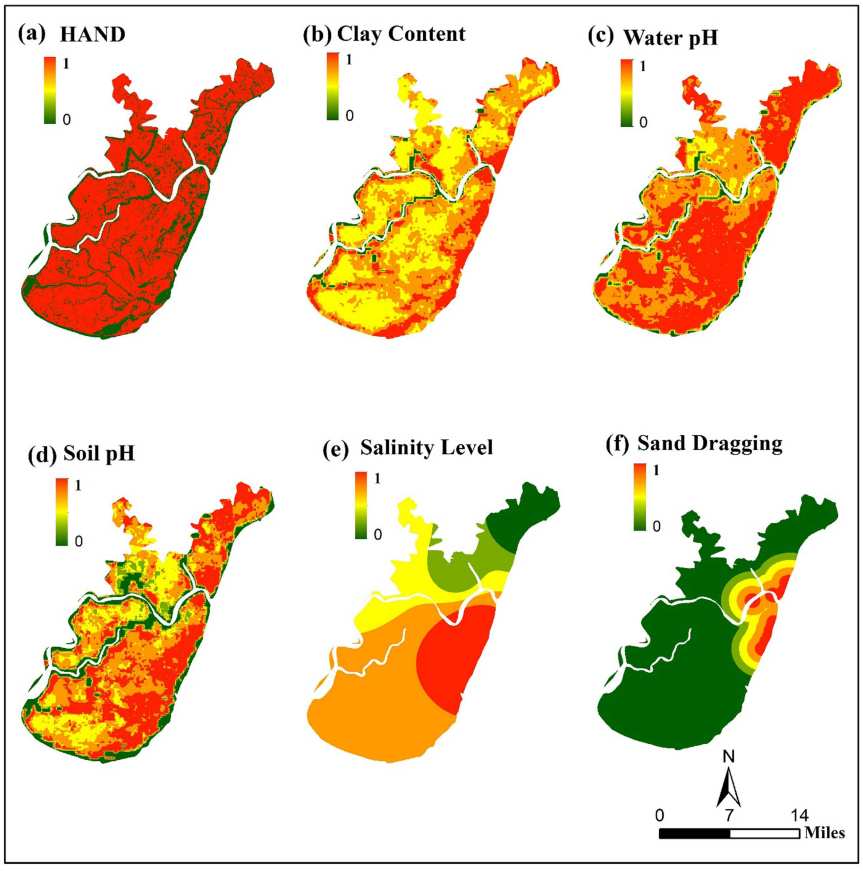


Figure 4. Additional hazard-related criteria used in the assessment: (a) HAND, (b) clay content, (c) water pH, (d) soil pH, (e) salinity level, and (f) sand dragging.

2.3.1.2. Cyclone hazard criteria. To assess cyclone-related hazard, multiple environmental and geospatial criteria were integrated into a multi-criteria analysis framework. Each criterion was categorized into five categories, ranging from Very Low (Rank 1) to Very High (Rank 5) hazard levels, based on threshold values derived from literature, satellite data, and regional cyclone studies. Below is a description of each variable.

Proximity to the coastline is a key factor in determining the level of exposure to cyclone hazards. Areas located less than 6 km from the coast fall into the very high risk category, as these zones are directly impacted by storm surges, coastal inundation, and strong winds associated with tropical cyclones. On the other hand regions more than 25 km inland (Very Low Risk) are comparatively shielded from the immediate destructive forces of cyclones (Paul 2009). Elevation is another critical topographic variable for cyclone-induced flood risk. Low-lying areas below 2 m are extremely vulnerable to tidal surges and cyclone-driven flooding, especially during high tide and storm events. Conversely, areas above 10 m in elevation are typically safe from such inundation, as their altitude provides natural protection (Table 3).

The slope of coastal terrain significantly influences how quickly water from storm surges or rainfall can drain. Gentle slopes (less than 1°) result in water stagnation and prolonged flooding, making these areas highly susceptible to cyclone-related hazards. Conversely, steep slopes (greater than 5°) facilitate faster drainage and minimize water accumulation, leading to a much lower flood risk. Aspect, or the direction a slope faces, also contributes to hazard exposure. East-facing slopes are more vulnerable to cyclonic winds from the Bay of Bengal which typically approach from the southeast or east placing these areas in the very high-risk category. On the other hand, northwest-facing slopes are naturally sheltered from direct cyclone impacts due to their orientation, placing them under very low risk (Table 3). Wind speed is a direct indicator of cyclone intensity, further defines hazard zones. Areas subjected to wind speeds above 125 km/h are at very high risk due to potential damage from flying debris, uprooted trees, and structural failure. In contrast, zones experiencing wind speeds below 80 km/h are significantly safer (Table 3) (Hoque et al. 2019a).

Table 3. Alternative ranking pattern of the hazards.

Hazard	Criteria layers	Units	Very low	Low	Moderate	High	Very high	Weight
Flood	Distance coast	km	>28	28–21	21–14	14–7	<7	0.11
	Clay content	%	0–5	5–12	12–20	20–29	29–39	0.08
	Coastal DEM	m	>11	11–8	8–5	5–2	<2	0.11
	Coastal slope	degree	>5.34	3.74–5.34	2.14–3.74	0.93–2.14	<0.93	0.07
	Stream density	km/km^2	<0.35	0.35–0.75	0.75–1.15	1.15–1.50	<1.5	0.06
	Distance stream	m	>1350	1350–951	950–551	550–251	>250	0.06
	Evaporation	mm/day	<0.75	0.751–1.85	1.851–3.25	3.251–4.95	<4.95	0.04
	Coastal flood	%	<5	5–10	108–16	16–22	<22	0.10
	LST	°C	<24	24–27	27–30	30–33	>33	0.03
	Hand	m	<2.2	2.21–5.6	5.61–9.8	9.81–14.5	<14.51	0.10
	River	m	>3935.66	3935.66–2640.38	2640.38–1569.28	1569.28–697.46	<697.46	0.06
	NDWI		<−0.10	−0.099 to 0.00	0.001–0.10	0.101–0.20	<0.20	0.04
	SPI		<−8.50	−8.50 to 2.50	−2.50 to 1.50	1.50–4.50	<4.50	0.04
	Rainfall	mm	<1686.20	1686.21–1943.92	1943.93–2229.54	2229.55–2515.16	<2515.17	0.10
	Aspect	degree	>135	135–90	90–45	45–0	<0	0.02
Cyclone	Distance coast	km	>25	18–25	12–18	6–12	<6	0.133
	Coastal DEM	m	>10	7–10	4–7	2–4	<2	0.166
	Coastal slope	degree	>5°	3°–5°	2°–3°	1°–2°	<1°	0.066
	Coastal aspect	Degree	Northwest-facing	Westfacing	South-facing	Southeast-facing	East-facing	0.016
	Wind speed	km/h	<80	80–95	95–110	110–125	>125	0.05
	Stream density	(km/km)	<0.4	0.4–0.8	0.8–1.2	1.2–1.6	>1.6	0.099
	Distance stream	m	>1500	1000-1500	500–1000	200–500	<200	0.016
	Evaporation	mm/day	<1	1–2	2–3	3–4	>4	0.017
	LST	°C	<26°C	26°C–28°C	28°C–30°C	30°C–32°C	>32°C	0.017
	River proximity	m	>3000	2000–3000	1000–2000	500–1000	<500	0.138
	Cyclone track	km	>10	7–10	4–7	2–4	<2	0.083
	SPI	Unitless	>1.5	0 to 1.5	−1 to 0	−2 to −1	<−2	0.017
	Rainfall	mm/year	<1600	1600–1900	1900–2200	>2500	>2500	0.072
	Stream density	km/km^2	<0.35	0.35–0.75	0.75–1.15	1.15–1.50	<1.50	0.099
	LST	°C	<24	24–27	27–30	30–33	>33	0.017
Salinity Intrusion	Coastal aspect	degree	>135	90–135	45–90	0–45	<0	0.01
	Coastal slope	degree	>5.34	3.74–5.34	2.14–3.74	0.93–2.14	<0.93	0.05
	Stream density	km/km^2	<0.35	0.35–0.75	0.75–1.15	1.15–1.50	<1.50	0.1
	LST	°C	<24	24–27	27–30	30–33	>33	0.04
	Cyclone track	km	>6	6–4.5	4.5–3	3–1.5	<1.5	0.02
	Distance coast	km	>28	28-21	21–14	14–7	>7	0.02
	Distance stream	m	<1351	950–1350	550–950	250–550	>250	0.08
	Evaporation	mm/day	<0.75	0.75–1.85	1.85–3.25	3.25–4.95	<4.95	0.03
	NDWI		>0.341	0.10–0.20	0.001–0.10	−0.099 to 0.00	>0	0.15
	Water pH		6.5–7.5	5.51–7.0	3.01–5.5	0–3	>8.5	0.01
	Soil pH		6.5–7.5	5.51–7.0	3.01–5.5	0–3	>8.5	0.01
	Coastal DEM	m	>11	11–8	8–5	5–2	<2	0.07
	Main river proximity	km	>8	8.0-6.0	6.0-4.0	4.0–2.0	>2	0.12
	Tide range	meters	<1.12	1.13–1.14	1.15–1.16	1.16–1.17	>1.17	0.03
	Storm surge	meter	<0.4	0.41—0.8	0.8–1.2	1.2–1.6	<1.6	0.03
	Salinity level	µS/cm	8861	8862—10,237	10,238–11,725	11,726—13,436	<13,437	0.2
	Rainfall	mm/year	<2200	2200–2550	2551–2900	2901–3250	>3250	0.05
Erosion	Stream density		<0.35	0.35–0.75	0.75–1.15	1.15–1.50	>1.50	0.1
	LST	km/km^2	<24	24–27	27–30	30–33	>33	0.015
	Coastal aspect	°C	>135	90–135	45–90	0–45	<0	0.01
	Coastal slope	degree	>5.34	3.74–5.34	2.14–3.74	0.93–2.14	<0.93	0.055
	Cyclone track	degree	>6	4.5–6	3–4.5	1.5–3	<1.5	0.015
	Distance coast	km	>28	21–28	14–21	7–14	<7	0.08
	Distance stream	km	>1351	951–1350	551–950	251–550	<250	0.09
	Evaporation	m	<0.75	0.75–1.85	1.85–3.25	3.25–4.95	>4.95	0.025
	NDWI	mm/day	>0.341	0.20–0.34	0.10–0.20	0.00–0.10	<0	0.045
	Water pH		6.5–7.5	5.51–7.0	3.01–5.5	0–3	>8.5	0.01
	Soil pH		6.5–7.5	5.51–7.0	3.01–5.5	0–3	>8.5	0.01
	Coastal DEM		>11	8–11	5–8	2–5	<2	0.07
	Tide range	m	<1.12	1.13–1.14	1.15–1.16	1.16–1.17	>1.17	0.09
	Main river proximity	m	>8	6.0–8.0	4.0–6.0	2.0–4.0	<2	0.11
	Storm surge	m	<0.4	0.41–0.8	0.8–1.2	1.2–1.6	>1.6	0.1
	Salinity level	m	<8861	8862–10,237	10,238–11,725	11,726–13,436	>13,437	0.035
	Rainfall	µS/cm	<2200	2200–2550	2551–2900	2901–3250	>3250	0.035
	Clay content	mm/year	<29	20–29	12–20	5–12	<5	0.055
	Sand dragging	%	>600	600–450	450-300	300–150	>150	0.055

Stream density plays a critical role in drainage efficiency. A high stream density (greater than 1.6 km/km^2) indicates poor infiltration, leading to surface water accumulation during cyclonic rainfall. In contrast, low stream density (less than 0.4 km/km^2) correlates with effective drainage and reduced waterlogging. Similarly, proximity to streams greatly influences flood vulnerability. Regions within 200 m of a stream are very highly exposed to flooding during heavy rainfall. While areas beyond 1500 m benefit from a natural buffer and face minimal risk. Evaporation rates also affect how long floodwaters persist after cyclones. High evaporation (>4 mm/day) supports faster water removal and reduces hazard duration, while low evaporation (<1 mm/day) leads to water stagnation and extended flooding (Table 3). Land Surface Temperature (LST), another indirect indicator, reflects surface characteristics; high LST values above 32°C often suggest impervious or urbanized surfaces that heighten runoff and flood risk. Lower LST values below 26°C are usually found in vegetated or water-absorbing zones, which can reduce cyclone-induced flood severity (Table 3).

Proximity to rivers influences cyclone-related flood hazards. Areas within 500 m of rivers are highly prone to flooding due to overflow from intense rainfall, while zones beyond 3000 m are less affected (Table 3). Cyclone track proximity is also critical; regions within 2 km of historical cyclone paths are highly vulnerable to wind and storm surge impacts, whereas locations beyond 10 km face minimal hazard (Table 3). The Standardized Precipitation Index (SPI) reflects rainfall anomalies, with values below −2 indicating intense wet spells linked to cyclone events and very high risk, while values above +1.5 represent drier conditions and lower flood exposure. Annual rainfall further strengthens this pattern, as regions receiving >2500 mm annually experience higher soil saturation and runoff during cyclones, increasing hazard intensity. In contrast, areas with <1600 mm rainfall are generally less prone to flooding and fall into the very low hazard category (Table 3) (Dasgupta et al. 2014).

2.3.1.3. Salinity intrusion hazard criteria. Salinity intrusion in coastal regions is a complex process influenced by various environmental, geographical, and hydrological parameters. For this study, a set of critical criteria were selected and categorized into salinity risk levels very low, low, moderate, high, and very high salinity risk classes.

Areas with very low stream density less than 0.35 km/km^2, tend to have reduced freshwater discharge, making them more susceptible to saline water creeping inland. In contrast, regions with very high stream density over 1.50 km/km^2 enjoy the protective effect of stronger freshwater flows that push back against salinity intrusion (Table 3). Similarly, cooler land surface temperatures (below 24°C) often mean lower evaporation rates, helping to maintain surface moisture and dilute salt levels. On the other hand, very high temperatures (above 33°C) speed up evaporation, leaving behind concentrated salts that worsen salinity conditions (Naher et al. 2011). The angle and slope of the coastline also matter. When the coast is oriented at angles greater than 135°, it tends to deflect saline flows away from land, offering natural protection. However, coastlines angled below 0° are more exposed and allow easier saltwater intrusion. Slope plays a similar role steep coastal slopes (more than 5.34°) encourage fast runoff, preventing saltwater from settling, while gentle or flat slopes (less than 0.93°) promote stagnation and allow saline water to seep inland (Table 3).

Cyclone activity has a surprising role in salinity levels. In places hit by more than six cyclones a year, the frequent storm induced freshwater flushes can actually reduce salt buildup. But where cyclones are rare fewer than 1.5 per year saltwater tends to linger longer, increasing the risk. Proximity to rivers and the sea adds another layer. Settlements within 7 km of a river or less than 250 m from the sea face much higher exposure to salinity due to tidal influences and the limited buffer distance (Akter et al. 2025). On the flip side, areas farther than 28 km from rivers or more than 1351 m from the sea are generally safer (Table 3). Evaporation also affects salinity low evaporation (under 0.75 mm/day) helps keep surfaces moist, while high evaporation (above 4.95 mm/day) leaves salt behind. The Normalized Difference Water Index (NDWI) helps detect water presence: high values (above 0.341) reflect good moisture conditions, while values below zero highlight dryness and elevated salinity risk (Hoque et al. 2019a). Water and soil pH also matter neutral pH levels (6.5–7.5) point to stable, low-salinity environments, while readings over 8.5 often suggest salinized and alkaline conditions (Table 3).

Other environmental and geographic variables deepen the picture. Coastal areas sitting more than 11 m inland from the high tide line tend to avoid direct saltwater impacts, but those within just 2 m are highly vulnerable. Similarly, sites within 2 km of major rivers often suffer more due to tidal push, while those over

8 km away enjoy relative safety. The natural tidal range is another indicator: smaller tidal ranges (under 1.12 m) limit inland salt movement, whereas larger ones (over 1.17 m) facilitate it. Storm surges also dramatically elevate salinity risk surges over 1.6 m can flood wide areas with saltwater, while smaller surges (under 0.4 m) have minimal effect (Table 3). Actual salinity levels, measured in conductivity (μS/cm), give a clear signal below 8861 μS/cm is considered safe, while levels above 13,437 μS/cm suggest serious salt stress (Ashrafuzzaman et al. 2022). Finally rainfall patterns influence salinity differently, less than 2200 mm of annual rain may help maintain a stable salt balance, but heavy rainfall above 3250 mm can alter soil chemistry, leach minerals, and ironically heighten salinity threats in vulnerable coastal region (Table 3).

2.3.1.4. Erosion hazard criteria. Categorizes erosion hazard levels based on various environmental and physical criteria are assigned values that fall under five erosion hazard levels Very Low, Low, Moderate, High, Very High.

Stream density plays a key role in erosion risk by indicating surface water flow. Areas with densities >1.5 km/km^2 experience concentrated runoff and channel erosion, making them very high hazard zones, while areas with <0.35 km/km^2 have lower water movement and reduced erosion potential (Sarwar and Woodroffe 2013). Land Surface Temperature (LST) also influences erosion; higher temperatures (>33°C) dry soils and weaken vegetation cover, increasing susceptibility, whereas lower LST (<24°C) supports moisture retention and vegetation stability, reducing erosion risk (Table 3). Coastal aspect further affects exposure, as areas facing 0°–45° are more exposed to winds and wave action, increasing erosion risk, while slopes facing away (>135°) are more protected (Sahoo and Bhaskaran 2018). Slope gradient also plays a role; gentle slopes (<0.93°) retain water and weaken soil cohesion, whereas steeper slopes (>5.34°) promote drainage and reduce erosion risk (Table 3).

2.3.2. Criteria for vulnerability mapping

For vulnerability analysis both social and Physical vulnerability criteria were considered.

2.3.2.1. Criteria for social vulnerability mapping. Social vulnerability reflects how population groups are affected by hazards such as floods, cyclones, salinity intrusion, and erosion (Rahman and Hossen 2024). The ability to prepare for, respond to, and recover from these events depends on socio-economic conditions and the physical environment. In this study, population density, literacy rate, population hotspots, female population, dependent population, and agricultural dependency were used as key indicators (Figure 5a–f). Population density was derived from union-level Bangladesh Bureau of Statistics (BBS) census data and represents demographic pressure within administrative units. In contrast, the population hotspot layer was generated from a gridded population dataset in Google Earth Engine and identifies localized clusters of high population concentration. Population density and population hotspots were both included because they represent complementary dimensions of social vulnerability. Population density reflects the overall demographic pressure across an area, whereas gridded population hotspots identify localized concentrations of people with potentially higher exposure to hazards. Therefore, both variables were retained to capture regional demographic patterns as well as localized population clustering, providing a more comprehensive assessment of social vulnerability. Since the variables were selected based on their distinct conceptual roles within the AHP framework, rather than solely on statistical relationships, a formal statistical redundancy assessment was not undertaken. Therefore, both variables were retained as they capture complementary aspects of social vulnerability rather than redundant information. High population density (>510 persons/km^2) increases vulnerability by complicating evacuation, straining emergency services, and overloading infrastructure, particularly in coastal and riverine areas. Low literacy (<44.76%) further limits awareness and response capacity, increasing risk. Vulnerability is amplified in population hotspots where >36.54% of residents live in high-risk zones such as floodplains or cyclone paths, while areas with <0.72% exposure face lower direct risk (Cutter et al. 2003).

Demographic characteristics also influence vulnerability. Areas with higher female population (>50.5%) may face added challenges due to caregiving roles and social barriers, whereas areas with lower proportions (<44.5%) may experience fewer constraints (Table 4). Higher literacy (>62.34%) enhances preparedness and response, while low-density areas (<360 persons/km^2) are easier to manage but may lack service

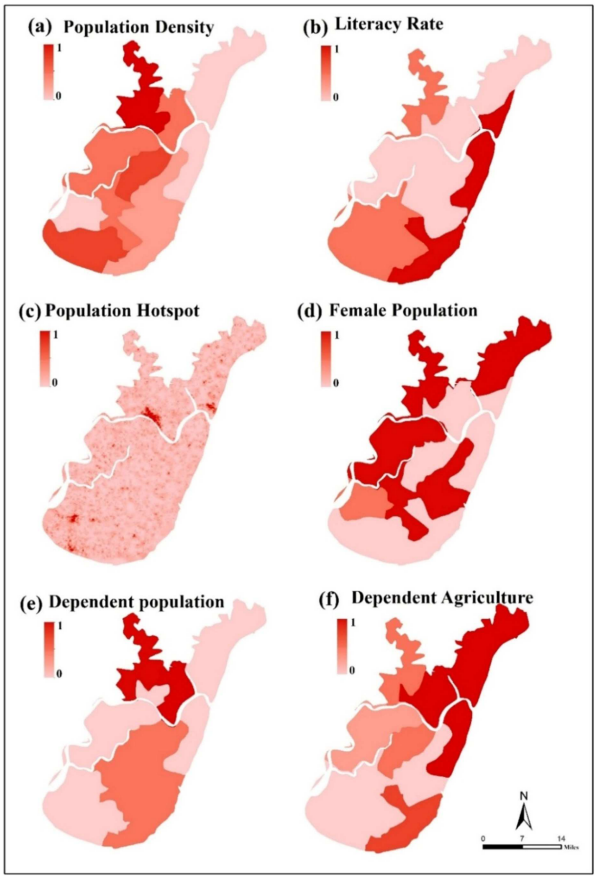


Figure 5. Social vulnerability mapping criteria used in the study: (a) population density, (b) literacy rate, (c) population hotspot, (d) female population, (e) dependent population, and (f) dependent agriculture.

access. A high dependent population (>49%) increases evacuation complexity, whereas lower dependency (<46%) supports resilience (Hoque et al. 2019b; Sweet and Rahman 2025). Similarly, strong reliance on agriculture (>8.5%) heightens vulnerability due to livelihood disruption, while lower dependence (<4.27%) improves adaptive capacity through income diversification (Table 4). Due to the unavailability of reliable, spatially explicit datasets at the Upazila level, certain commonly used social vulnerability indicators (e.g. disability status, household income, and livelihood diversification) were not included in this assessment. Therefore, only indicators with consistent and reliable spatial data were considered to ensure the robustness and comparability of the analysis. The omission of these variables may result in a partial representation of social vulnerability, particularly in areas where socioeconomic conditions differ substantially. Future studies incorporating these indicators, when reliable spatial data become available, could further improve the comprehensiveness and accuracy of vulnerability assessment.

2.3.2.2. Criteria for physical vulnerability mapping. Physical vulnerability was assessed using five key criteria: settlement density, NDVI, distance to settlement, coastal vegetation, and land use/land cover (Figure 6a–e). Each criterion was classified into five vulnerability levels—very low to very high—based on defined thresholds.

Settlement density is an important indicator of physical vulnerability to coastal hazards. Areas with low settlement density (0–150 km/km^2) are considered relatively less vulnerable because fewer people and assets are concentrated in these locations. In contrast, areas with very high settlement density (>1100 km/km^2) are classified as highly vulnerable because greater concentrations of settlements and built-up assets increase potential exposure to floods, cyclones, and salinity (Table 4). Although settlement density and distance to settlement are related to settlement distribution, they represent different spatial characteristics. Settlement density reflects the concentration of settlements within an area, whereas distance to settlement indicates the

Table 4. Alternative ranking of vulnerability and mitigation parameters.

Component	Criteria layers	Unit	Very low	Low	Moderate	High	Very high	Weight
Social vulnerability	Population density	(persons/km^2)	<360	360–410	411–460	461–510	>510	0.279
	Literacy rate	(%)	>62.34	56.49–62.34	50.63–56.48	44.77–50.62	<44.76	0.025
	Population hotspot	(%)	<0.72	0.72–3.67	3.67–13.17	13.17–36.54	>36.54	0.469
	Female population	(%)	<44.5	44.5–46.5	46.6–48.5	48.6–50.5	>50.5	0.117
	Dependent population	(%)	<46	46–47	47–48	48–49	>49	0.069
	Dependent agriculture	(%)	<4.27	4.27–5.68	5.69–7.09	7.10–8.5	>8.5	0.04
Physical vulnerability	Settlement density	(km/km^2)	0–150	151–400	401–750	751–1100	1101–1623.07	0.536
	NDVI	(Unitless)	0.201–0.2685	0.121–0.20	0.051–0.12	−0.019 to 0.05	−0.111 to 0.02	0.235
	Distance to settlement	(m)	1151–1570.62	851–1150	521–850	251–520	0–250	0.067
	Coastal vegetation	(%)	81–95	61–80	41–60	26–40	10–25	0.047
	LULC	(Categorical)	Water bodies	Forests	Agriculturalllands	Barren/ grassland	Settlements/ urban area	0.115
Mitigation	Cyclone centre proximity	(km)	<2	2–1.5	1.5–1	1–0.5	>0.5	0.439
	Health centre proximity	(km)	<6	4.5–6	4.5–3	3–1.5	>1.5	0.301
	Road distance	(km)	>0.6	0.6–0.45	0.45–0.3	0.3–0.15	<0.15	0.163
	Distance to coastal forest	(km)	>2	2–1.5	1.5–1	1–0.5	>0.5	0.097

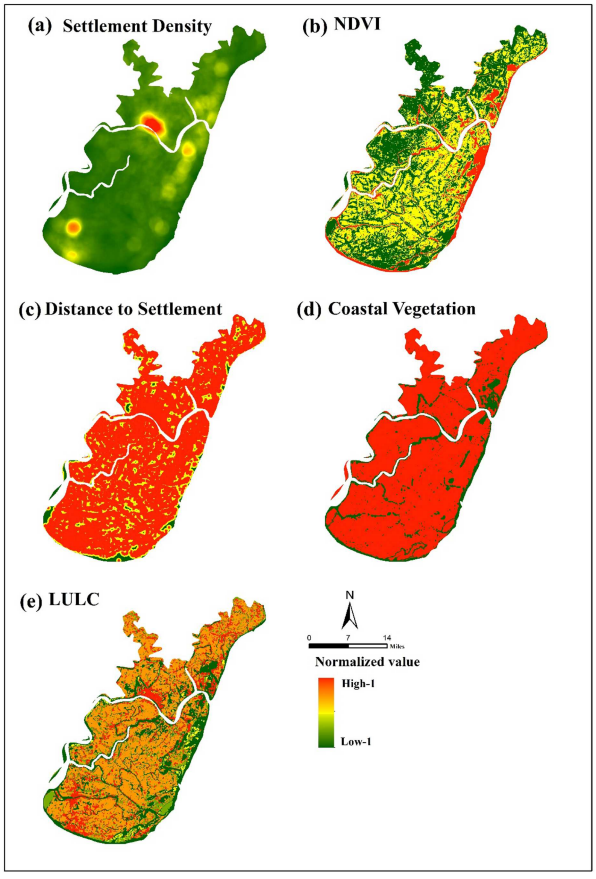


Figure 6. Physical vulnerability indicators used in the study: (a) settlement density, (b) NDVI, (c) distance to settlement, (d) coastal vegetation, and (e) land use/land cover (LULC).

proximity of locations to settlement areas. Therefore, both were retained as complementary criteria in the physical vulnerability assessment. Vegetation condition, represented by the Normalized Difference Vegetation Index (NDVI), also influences vulnerability. High NDVI values (0.201–0.2685) indicate healthier vegetation, which can reduce surface runoff and erosion and provide greater protection against environmental hazards (Murshed et al. 2022; Roy et al. 2022). Conversely, low or negative NDVI values (−0.111 to −0.02) indicate sparse or degraded vegetation and were therefore associated with higher vulnerability (Table 4).

Distance to settlement was considered to represent the spatial proximity of settlements and associated exposed assets. Areas farther from settlements (1151–1570 m) were assigned lower vulnerability, whereas locations within 250 m of settlements were classified as highly vulnerable because of their closer association with settlement areas (Table 4). Coastal vegetation provides an additional natural protective function; areas with 81%–95% vegetation cover were classified as very low vulnerability, whereas areas with sparse vegetation (10%–25%) were classified as very high vulnerability because of their limited protective capacity (Table 4) (Sahoo and Bhaskaran 2018; Hoque et al. 2019a).

Land use and land cover (LULC) further influence vulnerability. Natural landscapes such as forests and water bodies show low vulnerability due to their ecological functions. Agricultural lands exhibit moderate risk, whereas barren lands and grasslands are highly vulnerable due to weak soil stability. Urban areas, characterized by dense infrastructure and limited green space, are the most exposed and classified as very high risk, as they have limited capacity to absorb or withstand flooding, salinity, and storm surges (Hoque et al. 2019a).

2.3.3. Criteria for mitigation mapping

Spatial mitigation capacity was assessed using four key criteria: cyclone centre proximity, road distance, health centre proximity, and distance to coastal forest (Figure 7a–d). Each criterion was classified into five levels—very low to very high—based on threshold distances.

Proximity to cyclone centres strongly influences disaster response. Areas within 0.5 km of shelters have very high mitigation capacity due to quick access to safety, resources, and coordinated support. In contrast, locations beyond 2 km show very low mitigation capacity, as distance delays evacuation and limits access

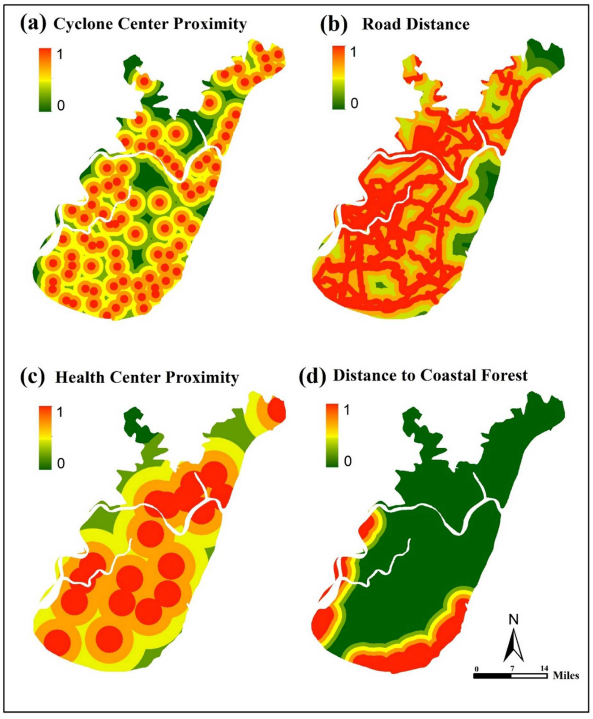


Figure 7. Mitigation capacity indicators used in the study: (a) cyclone shelter proximity, (b) road distance, (c) health centre proximity, and (d) distance to coastal forest.

to critical services (Table 4). Road accessibility is equally important; areas within 0.15 km of roads allow efficient evacuation and rapid delivery of emergency aid, resulting in very high mitigation capacity (Hoque et al. 2019a). Conversely, regions more than 0.6 km from roads face mobility constraints, reducing response efficiency and falling into the very low mitigation category (Table 4).

Access to health services is a key factor in disaster preparedness. Communities within 1.5 km of health centres have very high mitigation capacity due to rapid access to medical care during emergencies. In contrast, areas located more than 6 km away are highly vulnerable, with delayed or limited access to critical services, resulting in very low mitigation capacity (Table 4) (Hoque et al. 2019a).

Coastal forests also provide important natural protection. Areas within 0.5–1 km of these buffers exhibit high mitigation capacity, as vegetation reduces wind impact, flooding, and erosion. Locations 1–1.5 km away benefit moderately, while those 1.5–2 km away show low mitigation (IPCC 2022). Beyond 2 km, the protective effect is minimal, and such areas fall into the very low mitigation category, making them more exposed to coastal hazards (Table 4).

2.4. Alternative ranking and standardization criteria layers

We assessed all the selected criteria layers according to risk levels from one to five, with rank one indicating very low risk and rank five representing very high risk conditions (Tables 2 and 3) to streamline the multi-criteria decision-making process using the AHP method. Each criterion layer was standardized by converting its rank values to a linear scale (from 0 to 1). This standardization process was carried out using the following equation:

P = x-min/max—min

here, x stands for the cell value, while min and max represent the minimum and maximum values within each dataset, respectively. The variable p indicates the standardized score (Hoque et al. 2018).

2.5. Weighting the criteria using AHP

In this study, the widely used AHP was applied to assign weights to the hazard, vulnerability, and mitigation criteria for multi-hazard risk assessment. AHP was selected because it provides a systematic and transparent framework for integrating multiple spatial criteria, particularly in data-limited environments.

To improve the reliability of the weighting process, six nationally recognized experts were selected based on their expertise in GIS, remote sensing, disaster risk reduction, coastal hazard assessment, research experience, and knowledge of the study area and coastal Bangladesh. The panel included specialists from academic, governmental, and research institutions. Each expert independently evaluated the relative importance of the selected criteria using Saaty's nine-point pairwise comparison scale. The individual judgments were then reviewed and synthesized to obtain a final pairwise comparison matrix for each risk component (hazard, vulnerability, and mitigation). The resulting criterion weights were normalized so that the weights within each component summed to 1.0, with consideration given to expert judgments, published literature, and the relevance of each criterion to coastal multi-hazard assessment.

The consistency of the resulting pairwise comparison matrices was evaluated using the Consistency Ratio (CR) (Tables 2 and 3). A CR ≤ 0.10 was considered acceptable, indicating an acceptable level of consistency in the judgments. All matrices satisfied this criterion, supporting the reliability of the derived weights. The CR was calculated as follows:

$$\text{CR} = \text{Consistency Index/Random Index}$$

The random index (RI) represents the average consistency index generated randomly, while the consistency index (CI) can be calculated as follows:

$$\text{CI} = (\lambda \max - n)/(n - 1)$$

where, λ max represents the principal eigenvalue of the comparison matrix of the matrix plus and the matrix of order is denoted by n (Hoque et al. 2018).

2.5.1. Sensitivity analysis of AHP weights

To evaluate the robustness of the AHP-derived weights, an iterative sensitivity analysis was conducted during model development. The criterion weights were systematically adjusted across multiple scenarios (approximately 12–13 iterations) while maintaining logical consistency and acceptable consistency ratio (CR) values. The resulting hazard and risk maps from each iteration were visually compared to identify the most stable and geographically consistent spatial pattern. The final weight set presented in this study was selected because it produced the most realistic representation of observed hazard distribution and demonstrated stable spatial patterns across the tested scenarios.

2.6. Risk assessment

In this work, we used the weighted sum method to analyze the spatial layers associated with each of the three risk components. Based on this, we calculated separate indices for hazard, vulnerability, and Mitigation, and by applying the relevant weights to their contributing factors (Tables 3 and 4). These indices were subsequently grouped into five categories: very low, low, moderate, high, and very high. Next, we used ArcMap's raster calculator to generate a risk index by multiplying the component indices hazard and vulnerability and then dividing by the mitigation index to ensure consistency, we normalized the resulting risk values to a scale from 0 to 1 using. Finally, we categorized the normalized risk index into five levels very low, low, moderate, high, and very high based on the method applied by Hoque et al. (2018).

2.7. Risk validation

The reliability of the integrated multi-hazard risk model was evaluated using Receiver Operating Characteristic (ROC) curve analysis and the Area Under the Curve (AUC). A total of 110 georeferenced validation points were compiled from field investigations, local observations, and independent historical hazard information. During the field survey, representative locations affected by flooding, cyclones, salinity intrusion, and coastal erosion were visited and recorded using GPS. At each location, field evidence and observed hazard conditions were documented and subsequently compared with the corresponding predicted risk class. Independent historical hazard information obtained from government agencies and published records was also used to support the validation and cross-check field observations. A location was considered a positive observation when evidence of at least one of the four selected hazards was identified. Because the objective was to validate the integrated multi-hazard risk model, simultaneous occurrence of all four hazards at a single location was not required. The 110 validation points were spatially distributed across the study area to represent the observed hazard conditions and were overlaid on the predicted multi-hazard risk map. ROC–AUC analysis was then performed using the ArcSDM tool to evaluate the discriminatory performance of the model.

3. Result

3.1. Hazard mapping

Figure 8a–d illustrate the spatial variation and degree of hazard in the study region. The analysis incorporates four major hazards flooding, cyclones, salinity intrusion, and coastal erosion to generate a composite hazard index. The hazard map indicates that approximately 19.22% (84.97 km^2) of the study area falls within the very high hazard zone, while an additional 26.77% (118.44 km^2) is classified as a high hazard zone.

Most of these areas are located in the southeastern, southern, and southwestern parts of the study area, as indicated by the dominance of red and orange zones on the map. These areas lie in close proximity to the Bay of Bengal and active river networks, making them highly susceptible to storm surges, tidal flooding, saline water intrusion, and erosional processes. Moderate hazard zones account for 27.81% (123.03 km^2) of the total study area and are primarily located in the central part of the study area. Conversely, 18.22% (80.56 km^2) and 7.93% (35.05 km^2) of the area were categorized as low and very-low hazard zones, respectively. These zones are primarily located in the northern and northeastern parts of the

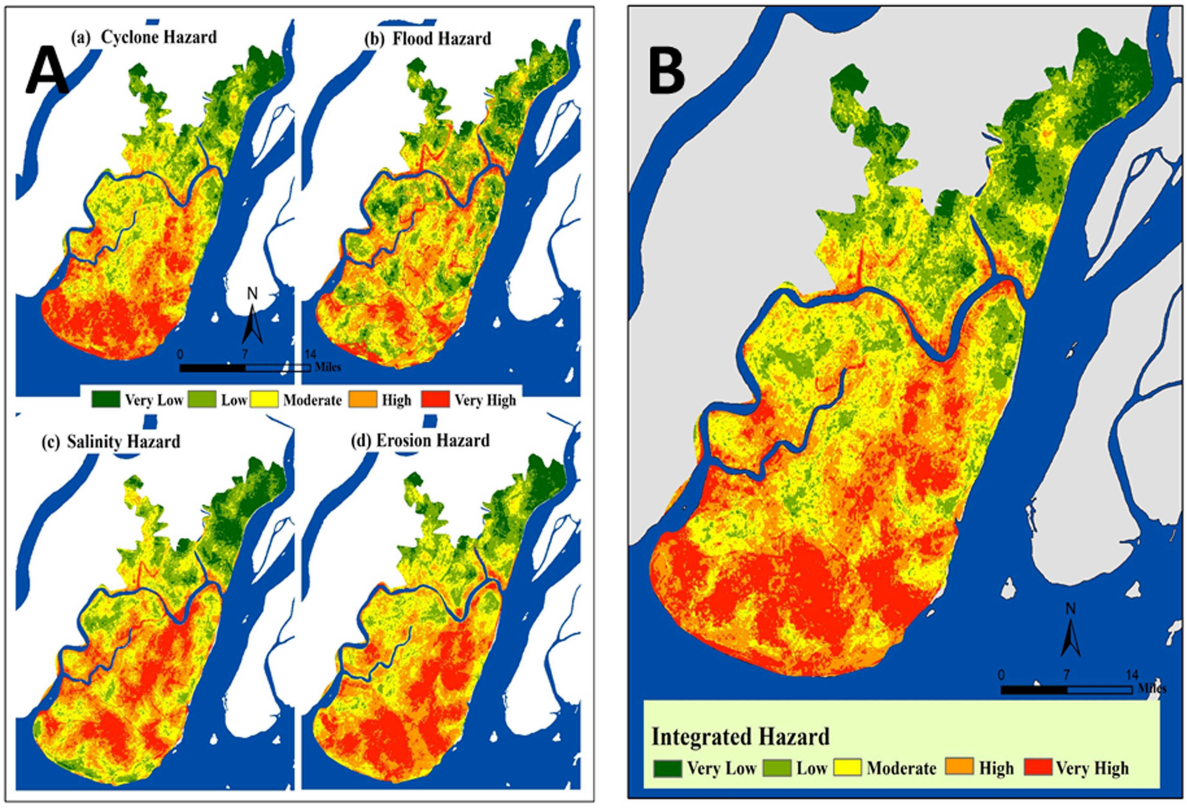


Figure 8. Spatial distribution of hazard components: in A (a) cyclone hazard, (b) flood hazard, (c) salinity intrusion hazard, (d) coastal erosion hazard, and (B) integrated hazard map.

study area and are associated with relatively higher elevations and greater distances from direct coastal influence. These areas experienced notably low hazard levels, mainly because they were less affected by strong winds, storm surges, and heavy rainfall.

3.2. Vulnerability mapping

3.2.1. Physical vulnerability mapping

Figure 9 illustrates the spatial distribution of physical vulnerability across the study area. Very high and high vulnerability zones cover 23.60% (104.77 km^2) and 21.07% (93.61 km^2), respectively, indicating that nearly half of the study area is highly vulnerable to multiple coastal hazards. These zones are mainly concentrated in the southern and central parts, with smaller portions in the north, highlighted in orange and red. Areas such as Dhulasar, Latachapli, and parts of Nilganj are located in low-lying regions near the Bay of Bengal and intersected by major rivers and tidal channels, increasing their susceptibility to tidal surges, cyclonic winds, and recurrent flooding.

The central-western part of Kalapara also exhibits high vulnerability, likely due to its proximity to erosion-prone riverbanks and inadequate drainage conditions. Moderate vulnerability zones account for 28.41% (126.19 km^2) of the area, mainly across the central inland region. Although not directly adjacent to the coast or major rivers, these areas remain exposed to seasonal flooding and salinity due to low elevation and limited protective infrastructure. Low and very low vulnerability zones cover 9.01% (40.01 km^2) and 17.91% (79.55 km^2), respectively, and are primarily located in the northern and northeastern parts, as well as some western and southwestern areas of Kalapara, including Tiakhali and parts of Dalbuganj, Dhulasar, and Latachapli.

3.2.2. Social vulnerability mapping

Figure 10b shows the spatial distribution of socio-economic vulnerability to multiple hazards across the study area. Very high and high vulnerability zones account for 4.14% (18.34 km^2) and 23.58%

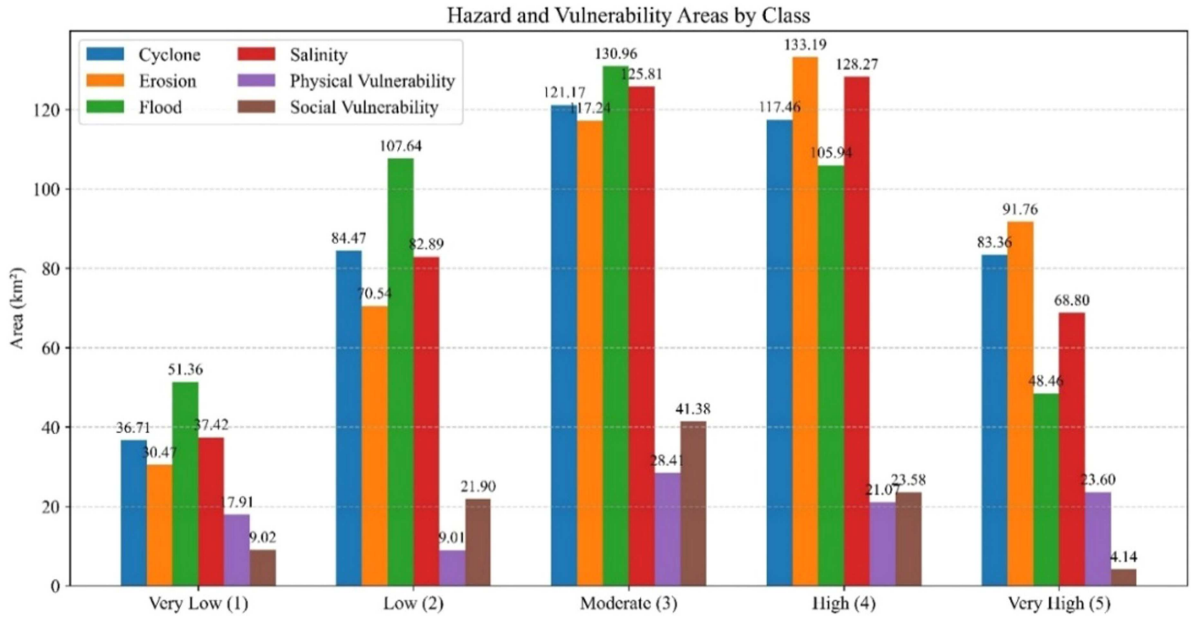


Figure 9. Spatial distribution of hazard and vulnerability classes in Kalapara Upazila.

(104.51 km^2), respectively. Very high vulnerability is concentrated in the southern and southwestern unions such as Dhulasar, Latachapli, and parts of Mahalchhari, while high vulnerability areas extend across central and southeastern regions, including Baliatali and Chakamaiya. These areas likely face multiple socio-economic challenges, including poverty, limited access to education and healthcare, inadequate housing, and weak institutional support.

Moderate vulnerability covers 41.38% (183.39 km^2), mainly in eastern and some southern areas. Low and very low vulnerability zones account for 21.9% (97.03 km^2) and 9.02% (39.99 km^2), respectively, and are primarily located in the north-central and northeastern parts, such as Tiakhali, where better infrastructure, higher literacy, and diversified livelihoods enhance resilience. A small area in the far north, likely including Dalbuganj, exhibits very low vulnerability.

3.2.3. Overall vulnerability mapping

Figure 10a–c illustrates the spatial distribution of integrated vulnerability to multiple coastal hazards across the study area. Very high and high vulnerability zones cover 10.38% (50.72 km^2) and 32.36% (158.04 km^2), respectively, mainly in the southern, southwestern, and central regions. Areas such as Dhulasar, Latachapli, and parts of Chakamaiya are highly exposed to both environmental hazards and socio-economic pressures.

Moderate vulnerability accounts for 25.63% (125.28 km^2), primarily in central and some southern interior areas, where either physical or social vulnerability is present but not at extreme levels. In contrast, low and very low vulnerability zones cover 17.7% (86.5 km^2) and 4.44% (21.72 km^2), respectively, mainly in the northern and northeastern parts, including Dalbuganj and Tiakhali. These areas represent relatively safer zones, where lower environmental exposure and improved socio-economic conditions enhance overall resilience.

3.3. Mitigation capacity mapping

Figure 11 illustrates the overall mitigation capacity of the study area in response to multiple natural hazards. Areas with high mitigation capacity represent locations where effective disaster risk reduction measures and supporting infrastructure are available to minimize hazard impacts.

The mitigation map shows that very high and high mitigation capacity areas cover 24.21% (119.21 km^2) and 14.15% (69.66 km^2) of the study region, mainly in the southwestern and southern parts of Kalapara. Key areas include Dhulasar, Latachapli, Nilganj, Tiakhali, Chakamaiya, and parts of Mithaganj, benefiting

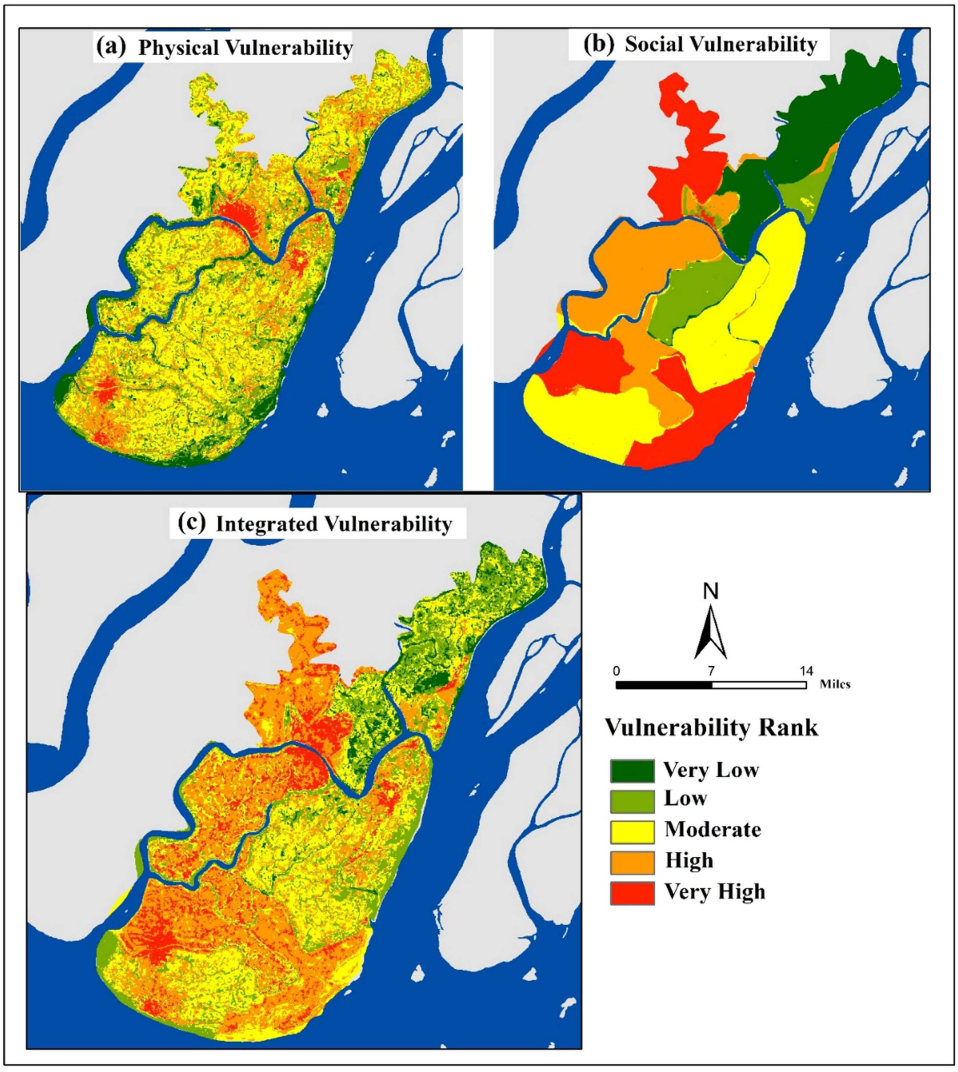


Figure 10. Vulnerability assessment maps: (a) physical vulnerability, (b) social vulnerability, and (c) integrated vulnerability.

from improved infrastructure, cyclone shelters, road access, and community preparedness. Moderate mitigation capacity accounts for 39.46% (194.26 km^2), primarily in central and mid-northern areas such as Kalapara Sadar and parts of Mithaganj and Tiakhali, where basic infrastructure exists but some vulnerability remains. Low and very low mitigation zones cover 5.9% (29.48 km^2) and 6.18% (30.44 km^2), mainly in the northern and northeastern regions, including Dhankhali, Lalua, and Chakamaiya, where limited access to emergency services and infrastructure increases exposure to climate hazards.

3.4. Risk mapping

The integrated risk map of Kalapara Upazila (Figure 12) reveals significant spatial variability in climate-related risks. Very high- and high-risk zones cover 16.08% (71.04 km^2) and 34.03% (150.36 km^2) of the study area, respectively., mainly concentrated in the southern and southwestern unions such as Lalua, Khaprabhanga, Baliatali, and parts of Latachapli and Mithaganj. These areas are located near the coastline and exhibit high exposure, high vulnerability, and comparatively limited mitigation capacity, making them highly susceptible to multiple coastal hazards. Moderate risk accounts for 27.66% (122.16 km^2), indicating substantial exposure across unions like Chakamaiya, Paurashava, Dalbuganj, and Dhulashar. In contrast, low and very low risk zones cover 14.25% (62.94 km^2) and 7.76% (34.25 km^2), primarily in the northern and central regions, including Tiakhali, Nilganj, and parts of Chakamaiya and Dhankhali. These areas

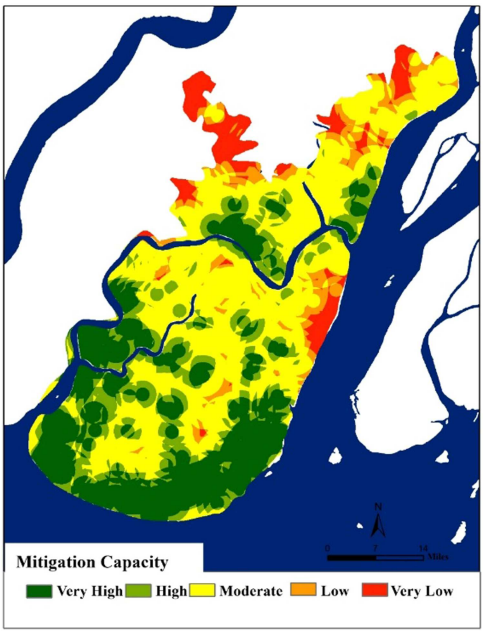


Figure 11. Spatial distribution of mitigation capacity across the study area.

benefit from higher elevation, reduced salinity, better infrastructure, and effective mitigation practices. Notably, Dhankhali shows extensive very low-risk zones, making it one of the most resilient unions in the Upazila.

3.5. Validation of risk map

To evaluate the predictive performance and reliability of the developed multi-hazard risk model for Kalapara Upazila, a Receiver Operating Characteristic (ROC) curve was generated using validation data (Figure 13). In this analysis, the ROC curve illustrates the ability of the model to distinguish between high-risk and low-risk zones for multiple hazards, including flood, cyclone, salinity intrusion, and erosion.

The area under the curve (AUC) value was 0.91, indicating excellent predictive accuracy and model performance. This AUC value signifies that the model correctly predicts hazard-prone areas 91% of the time, demonstrating strong discriminatory power. The curve sharply rises toward the top-left corner of the graph, reflecting a high rate of true positives with minimal false positives. For comparison, a diagonal black dashed line was included to represent a random guess model (AUC = 0.5), while a blue dashed line depicted a perfect classification scenario (AUC = 1.0). The proximity of the actual ROC curve to the ideal line, and its significant distance above the random guess line, confirms the robustness and practical applicability of the model. Furthermore, the consistency ratios (CR) for the hazard (0.008), vulnerability (0.0105), and mitigation (0.062) pairwise comparison matrices were all below the acceptable threshold (CR < 0.10), confirming the reliability and consistency of the expert judgments. These results validate that the selected indicators and weighting methods using the analytic hierarchy process (AHP) are effective in identifying and classifying multi-hazard risk zones in Kalapara. The high model accuracy supports its potential use in local-level disaster risk reduction (DRR), resource allocation, and climate change adaptation planning. The iterative adjustment of AHP weights resulted in only minor changes in the spatial distribution of hazard and risk classes. High- and very-high-risk zones remained concentrated in the southern coastal part of Kalapara throughout the tested iterations. The final weight combination was therefore considered robust for representing the overall spatial pattern of multi-hazard risk.

4. Discussion

Although this study emphasizes resilience implications, the proposed framework is designed for multi-hazard risk assessment rather than direct resilience assessment. The resulting risk maps support resilience

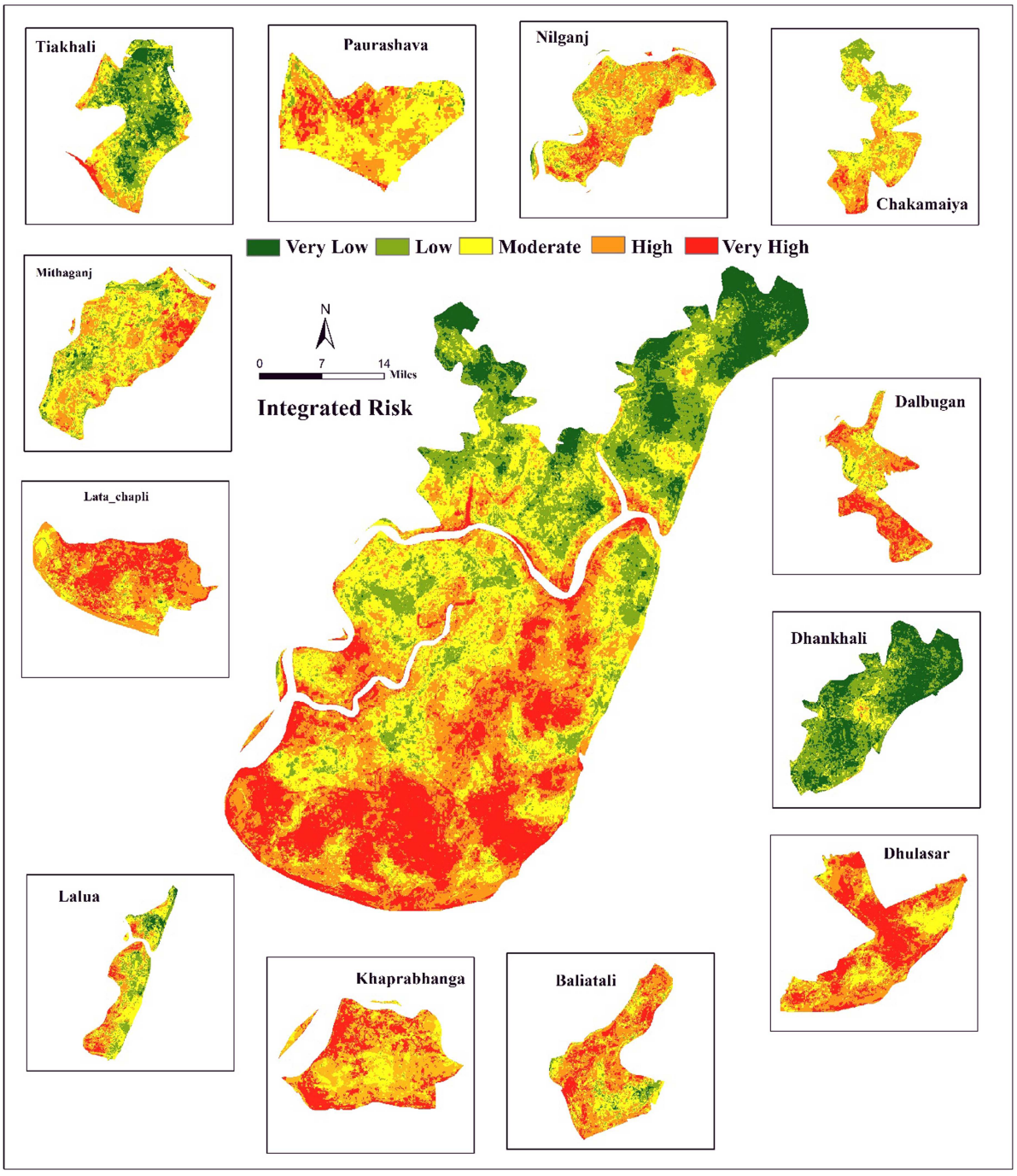


Figure 12. Integrated multi-hazard risk map and union-wise risk distribution of Kalapara Upazila.

planning by identifying priority areas for disaster risk reduction, mitigation, and climate adaptation. Coastal regions are increasingly exposed to multiple climate-related hazards, including cyclones, floods, salinity intrusion, and coastal erosion, which threaten livelihoods, infrastructure, and ecosystems. These challenges are particularly pronounced in Kalapara Upazila, where integrated multi-hazard risk assessment is essential for effective disaster risk reduction and resilience planning. Previous studies have largely examined either physical or socio-economic dimensions using broad-scale or grid-based approaches (Fitton et al. 2019). By integrating hazard, vulnerability, and mitigation capacity within a GIS-based multi-criteria decision-making framework, this study provides a more comprehensive assessment of local-scale multi-hazard risk. The concentration of high-risk zones in the southern and southwestern parts of

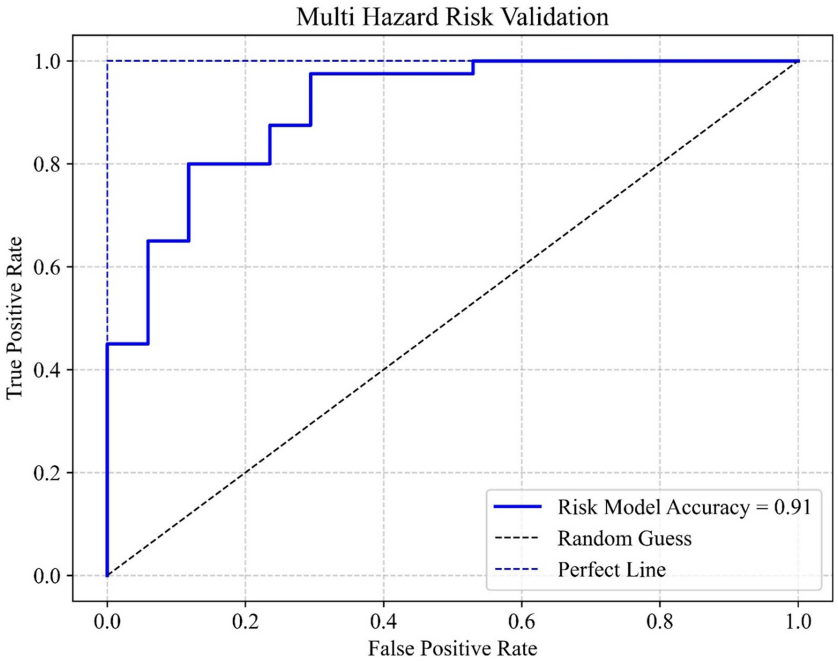


Figure 13. Receiver operating characteristic (ROC) curve and area under the curve (AUC) for validation of the multi-hazard risk model.

Kalapara reflects the combined influence of low-lying terrain, coastal proximity, and greater exposure to multiple hazards, whereas the northern and northeastern areas exhibit comparatively lower risk. These spatial patterns are consistent with previous studies conducted in coastal Bangladesh (Hoque et al. 2019; Khanam et al. 2020; Ahmed et al. 2025), further supporting the robustness of the present assessment. However, minor differences from previous studies may be attributed to variations in study area characteristics, spatial resolution, dataset acquisition periods, and the selection of evaluation criteria. Furthermore, the integration of hazard, vulnerability, and mitigation capacity within a single GIS–AHP framework provides a more comprehensive local-scale assessment than studies focusing on individual hazards or specific risk components. Beyond identifying high-risk areas, the proposed GIS–AHP framework also supports resilience-oriented planning by identifying priority locations for disaster preparedness, adaptive infrastructure, ecosystem conservation, and targeted risk reduction. These findings can assist local decision-makers in strengthening long-term climate resilience in coastal Bangladesh (Islam et al. 2016; Chowdhury et al. 2026).

This spatial pattern is mainly associated with the low-lying topography, close proximity to the Bay of Bengal, dense river networks, and frequent exposure to storm surges and saline water intrusion. Although the hazards were assessed separately, they are closely interconnected within coastal systems. In Kalapara, cyclones often trigger storm surges and flooding, which facilitate salinity intrusion into land and water resources. Recurrent saline inundation can degrade vegetation cover and soil quality, thereby increasing erosion risk. Likewise, erosion may weaken natural and engineered coastal defences, increasing vulnerability to future flood and cyclone events. These interactions indicate that high-risk zones are affected by cascading and compounding hazard processes rather than by isolated hazards. The intensity and spatial extent of these cascading effects may differ from previous studies due to variations in local geomorphological settings, hydrological conditions, and mitigation capacity.

The present findings are consistent with previous studies identifying coastal proximity and low elevation as major determinants of multi-hazard vulnerability in southern Bangladesh (Hoque et al. 2019). Similarly, Ahmed et al. (2025) demonstrated that the interaction between climatic hazards and socio-economic vulnerability substantially increases disaster risk in coastal communities, while Mukul et al. (2019) highlighted the high vulnerability of coastal settlements and ecologically sensitive environments to cyclones, flooding, and sea-level rise. Consistent with these studies, the highest risk levels in the

present assessment were concentrated in Lalua, Khaprabhanga, Baliatali, Latachapli, and Mithaganj (Figure 12), reflecting the combined influence of coastal proximity, low elevation, and limited mitigation capacity. The consistency with previous studies may be explained by the similar low-lying deltaic setting and exposure to interacting coastal hazards, while differences in risk patterns may arise from variations in spatial scale, criteria selection, and datasets. Unlike previous studies that primarily focused on individual hazards or specific risk components, this study integrates hazard, vulnerability, and mitigation capacity within a single GIS–AHP framework, providing a more comprehensive and locally relevant representation of multi-hazard risk.

The interacting coastal hazards identified in this study contribute to prolonged flooding, increased soil salinity, and land-use changes, including the expansion of salt-tolerant agriculture (Ahmed et al. 2025). Low-lying areas near the Bay of Bengal and adjacent river systems are particularly susceptible to flooding (Hoque et al. 2019), while climate change is expected to further intensify these risks (Bromwich et al. 2013). Human activities, particularly shrimp farming, have also accelerated salinity intrusion and environmental degradation (Roy et al. 2022). Although local communities have adopted adaptation measures such as rainwater harvesting and elevated housing (Habiba et al. 2012), broader resilience strategies remain insufficient (Hossain and Majumder 2018). The identified high-risk zones therefore provide valuable guidance for prioritizing adaptation measures, strengthening critical infrastructure, restoring coastal ecosystems, and promoting community-based resilience. Differences from previous assessments may reflect variations in local environmental conditions, data sources, and assessment approaches. The integrated consideration of multiple interacting hazards in the present study therefore provides a more context-specific basis for prioritizing resilience and adaptation measures in Kalapara.

The study also has important implications for disaster risk management and spatial planning. The spatially explicit identification of high-risk areas enables decision-makers to prioritize investments in cyclone shelters, evacuation infrastructure, ecosystem restoration, and climate-resilient land-use planning (Figures 11 and 12). Moreover, the integrated GIS–AHP framework provides a practical and transferable approach for multi-hazard risk assessment that can be adapted to other hazard-prone coastal regions of Bangladesh and similar deltaic environments. In particular, the identified high-risk zones can support location-specific prioritization of disaster preparedness and adaptation interventions rather than uniform resource allocation across the study area. A formal quantitative sensitivity analysis was not performed; however, an iterative sensitivity assessment was conducted during model development to examine the stability of the assigned weights. Nevertheless, this robustness reflects stability under the tested weight variations and does not eliminate uncertainty associated with expert judgment and the input datasets; therefore, future studies may apply more comprehensive approaches such as Monte Carlo simulation or global sensitivity analysis to quantify uncertainty more systematically.

This study has several limitations. Due to the limited availability of high-resolution datasets, Sentinel-2 imagery and a 30 m DEM were used, and some variables, including drainage density, early warning coverage, and groundwater table, could not be incorporated. The datasets also represent different acquisition periods (2021–2024), which may introduce minor uncertainty but are unlikely to substantially affect the overall assessment. Furthermore, the conventional AHP framework assumes independence among evaluation criteria and does not explicitly account for uncertainties in expert judgments or spatial datasets. Sensitivity analysis was not performed and should be considered in future studies to evaluate the robustness of the assigned weights. To improve the robustness of the weighting scheme, an iterative sensitivity assessment was conducted during model development by testing multiple weight combinations (approximately 12–13 iterations). The final AHP weights were selected based on the stability and consistency of the resulting hazard patterns while maintaining acceptable consistency ratio (CR) values. These iterative adjustments produced only minor variations in the overall spatial distribution of hazard classes, indicating that the final model is relatively robust to moderate changes in expert-assigned weights. Despite these limitations, the proposed model demonstrated strong predictive performance (AUC = 0.91; Figure 13), indicating that the integrated GIS–AHP framework is sufficiently robust for regional-scale multi-hazard risk assessment. Future research should incorporate higher-resolution elevation data (e.g. LiDAR), additional hazard and mitigation indicators, multi-temporal datasets, more comprehensive quantitative uncertainty assessment (e.g. Monte Carlo simulation or percentage-based sensitivity analysis) and uncertainty and sensitivity analyses. Comparative evaluations with advanced approaches, such as

Fuzzy AHP, ANP, Fuzzy Logic, ensemble machine learning, and deep learning models could further clarify methodological differences and uncertainty and help determine whether alternative approaches produce substantially different spatial risk patterns.

5. Conclusion

This study developed and validated an integrated GIS-, remote sensing-, and AHP-based multi-hazard risk assessment framework for Kalapara Upazila, one of the most hazard-prone coastal regions of Bangladesh. The resulting multi-hazard risk map showed that approximately half of the study area falls within high- to very high-risk zones, with the greatest risk concentrated in the southern and southwestern coastal unions. These findings improve the understanding of the spatial distribution of multi-hazard risk and identify priority areas for disaster risk reduction and resilience planning.

The study contributes to the existing literature by integrating hazard, vulnerability, and mitigation capacity within a single geospatial framework at the Upazila scale and validating the resulting risk map (AUC = 0.91), thereby providing a reliable and locally relevant tool for evidence-based decision-making. The generated risk map can support policymakers and disaster management agencies in prioritizing mitigation measures, climate adaptation, land-use planning, infrastructure development, and resource allocation in vulnerable coastal communities. Unlike many previous regional studies that primarily focused on individual hazards or coastal vulnerability assessments, this study provides an integrated multi-hazard geospatial framework that simultaneously incorporates hazard, vulnerability, and mitigation capacity, offering a more comprehensive basis for resilience-oriented planning and decision-making. Future research should incorporate higher-resolution datasets, additional hazard and mitigation indicators, uncertainty and sensitivity analyses, and advanced approaches such as Fuzzy AHP, ANP, and machine learning to further improve multi-hazard risk assessment and resilience planning. The proposed framework provides a practical tool for resilience-oriented coastal planning by supporting disaster risk reduction, climate adaptation, and evidence-based decision-making. Compared with previous coastal vulnerability assessments, the integration of hazard, vulnerability, and mitigation capacity provides more comprehensive support for resilience-building in hazard-prone coastal regions.

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Author contributions

CRediT: **Sharmin Akter:** Conceptualization, Formal analysis, Methodology, Software, Supervision, Validation, Visualization, Writing – original draft; **Muhammad Al-Amin Hoque:** Conceptualization, Methodology, Resources, Software, Supervision, Validation, Writing – review & editing; **Mehedi Hasan:** Conceptualization, Methodology, Resources, Software, Visualization, Writing – review & editing; **Biswajeet Pradhan:** Conceptualization, Supervision, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The datasets are available from the corresponding author upon reasonable request.

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